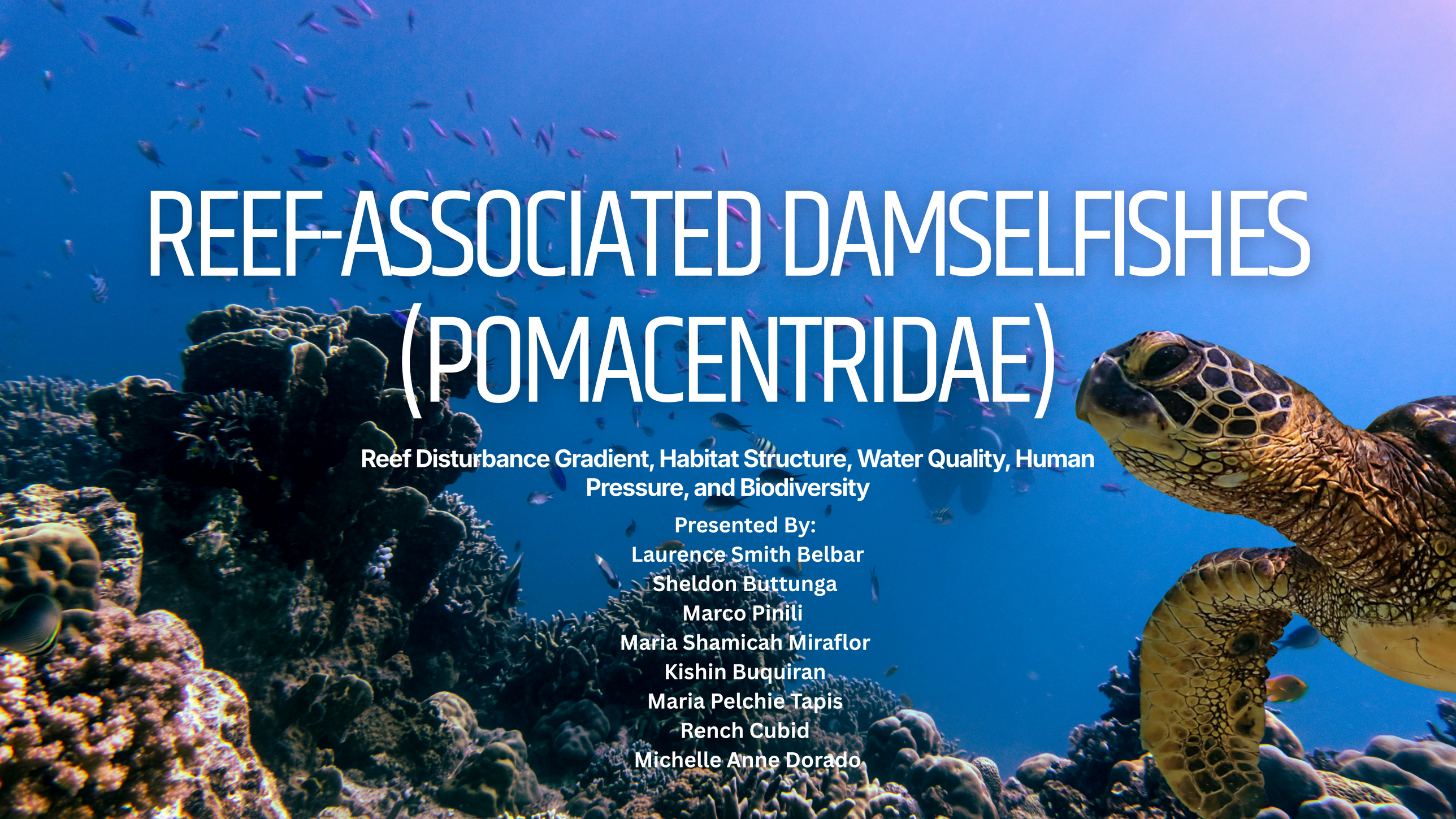


REEF-ASSOCIATED DAMSELFISHES (POMACENTRIDAE)

Reef Disturbance Gradient, Habitat Structure, Water Quality, Human Pressure, and Biodiversity

Presented By:

Laurence Smith Belbar

Sheldon Buttunga

Marco Pinili

Maria Shamicah Miraflor

Kishin Buquiran

Maria Pelchie Tapis

Rench Cubid

Michelle Anne Dorado

INTRODUCTION

- COASTAL REEFS ARE AFFECTED BY HUMAN ACTIVITIES AND ENVIRONMENTAL PRESSURES, SUCH AS POLLUTION, FISHING, TOURISM, AND COASTAL DEVELOPMENT.
- THESE PRESSURES CAN ALTER REEF HABITATS AND WATER CLARITY.
- THE STUDY EXAMINES REEF STRUCTURE, BENTHIC COMPOSITION, TURBIDITY, AND NEARBY HUMAN POPULATION PRESSURE ACROSS TEN PROPOSED SITES IN CENTRAL VISAYAS, PHILIPPINES.
- **ALLEN CORAL ATLAS:** PROVIDES REEF HABITAT, GEOMORPHIC CLASSES, AND TURBIDITY DATA.
- **WorldPop:** Provides population-density estimates.
- **GBIF:** Provides damselfish occurrence records to summarize species richness and data availability.

OBJECTIVES

- Describe reef geomorphic structure and mapped benthic composition within identical 20-km-radius buffers.
- Compare annual median turbidity and mean valid-land-pixel population density among the ten sites.
- Summarize GBIF damselfish occurrence records, observed species richness, genus richness, and family richness.
- Explore associations between observed richness and environmental predictors using Spearman correlations, and a small regression/GLM.
- Evaluate data limitations, including uneven GBIF sampling, spatial overlap, remote-sensing classification, and the small number of sites.

MATERIALS AND DATA SOURCES

1. **Allen Coral Atlas:** Reef structure, benthic composition, and turbidity.

2. **WorldPop:** 2020 population-density data.

3. **GBIF:** Damselfish occurrence records.

4. **R packages:** Spatial analysis, data cleaning, and statistics.

METHODS

1. **Spatial Design**

- Create 20-km-radius buffers for ten sites using R.
- Export GeoJSON files and check buffer overlaps.

2. **Allen Coral Atlas**

- Measure reef area, geomorphic classes, and benthic composition.
- Convert turbidity values to FNU by dividing by 10.

3. **WorldPop**

- Calculate mean population density using valid land pixels.
- Use population density as a proxy for human pressure.

METHODS

4. **GBIF**
- Retrieve damselfish records from 2000 onward.
- Filter inaccurate and duplicate records.
- Summarize records and taxonomic richness.

5. **Statistical Analysis**
- Use graphs, Spearman correlation, and regression or Poisson GLM.
- Interpret results cautiously due to the ten-site sample size.

RESULTS AND DISCUSSIONS

GBIF

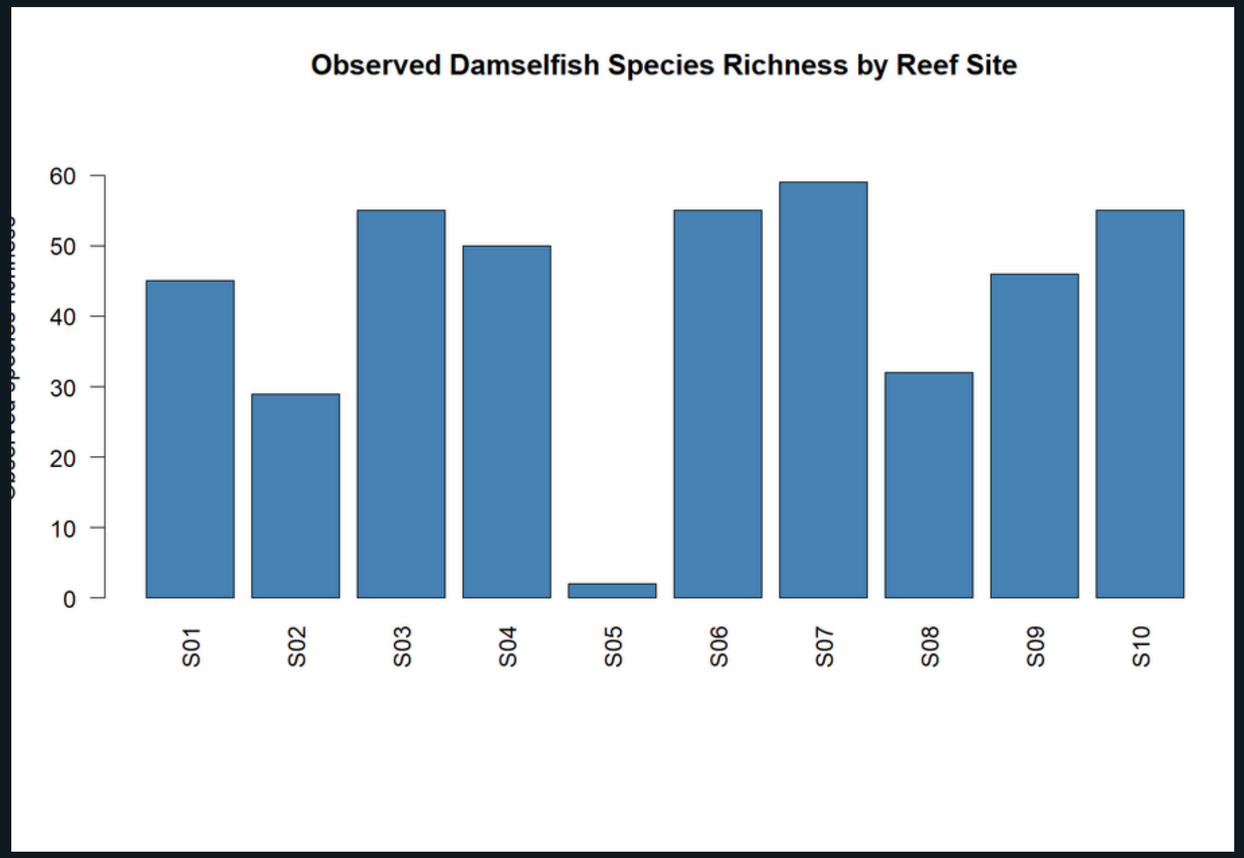


FIGURE 1. OBSERVED SPECIES RICHNESS.

- This graph shows the observed number of damselfish species at each of the 10 selected reef sites. To create it, I used RStudio to process the GBIF occurrence records, clean the data, group the records by site and species, and count the distinct species recorded at each site. Panglao Island Coastal Reef had the highest observed richness, with 59 species, while Danajon Bank had the lowest, with 2 species.
- How I did it in RStudio:
- Imported the GBIF occurrence data.
- Cleaned the data and assigned records to the selected reef sites using 20-km buffers.
- Grouped the records by site and species.
- Counted the distinct recorded species for each site.
- Generated the bar graph using R.

GBIF

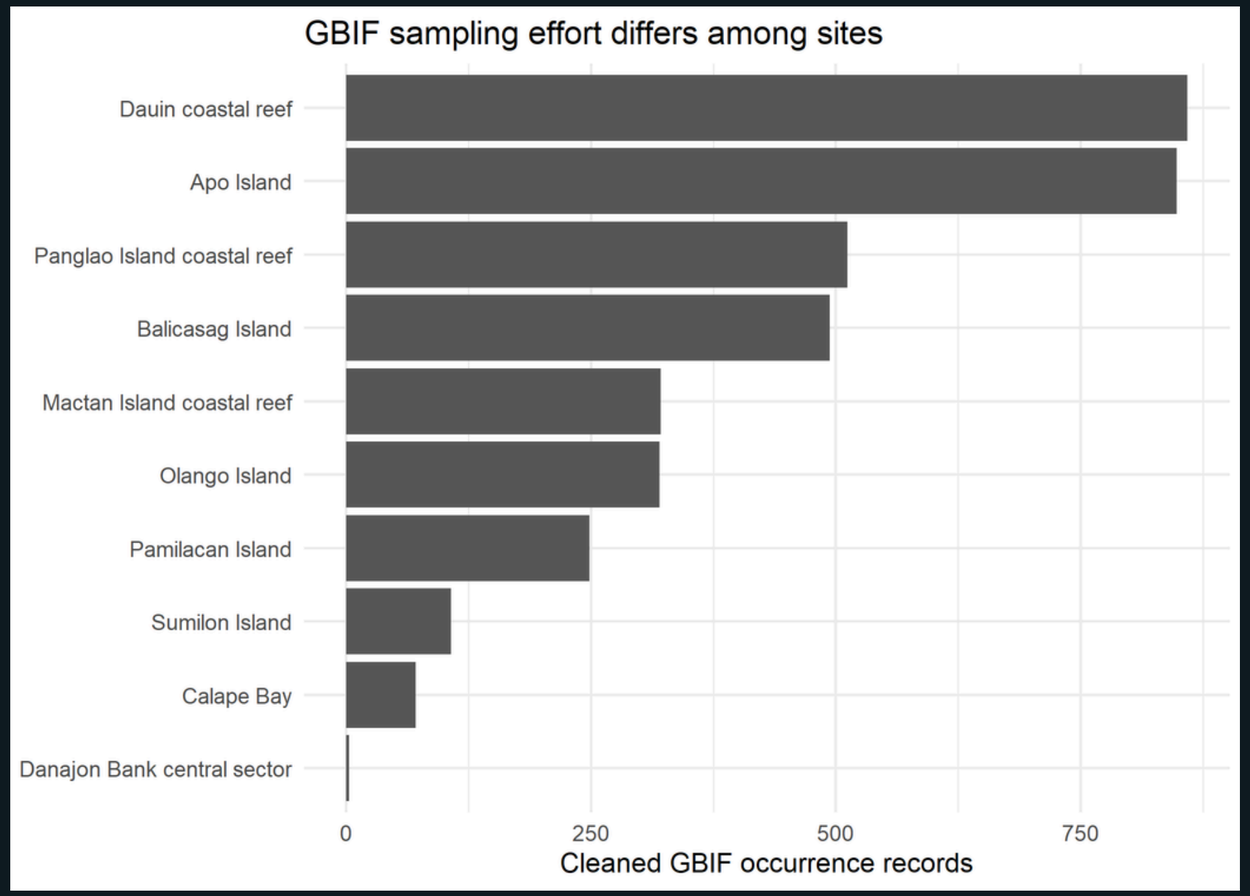


FIGURE 2: GBIF_RECORDS_BY_SITE

- This graph shows the number of GBIF occurrence records available for each selected reef site. Using RStudio, I counted the records assigned to each site and plotted the totals. Dauin Coastal Reef had 859 records, while Apo Island had 848. Danajon Bank had only 3 records, showing that the amount of available data differed greatly among sites.
- How I did it in RStudio:
- Used the cleaned GBIF occurrence dataset.
- Grouped the records by reef site.
- Counted the number of occurrence records per site.
- Created a bar graph to compare the totals.

GBIF

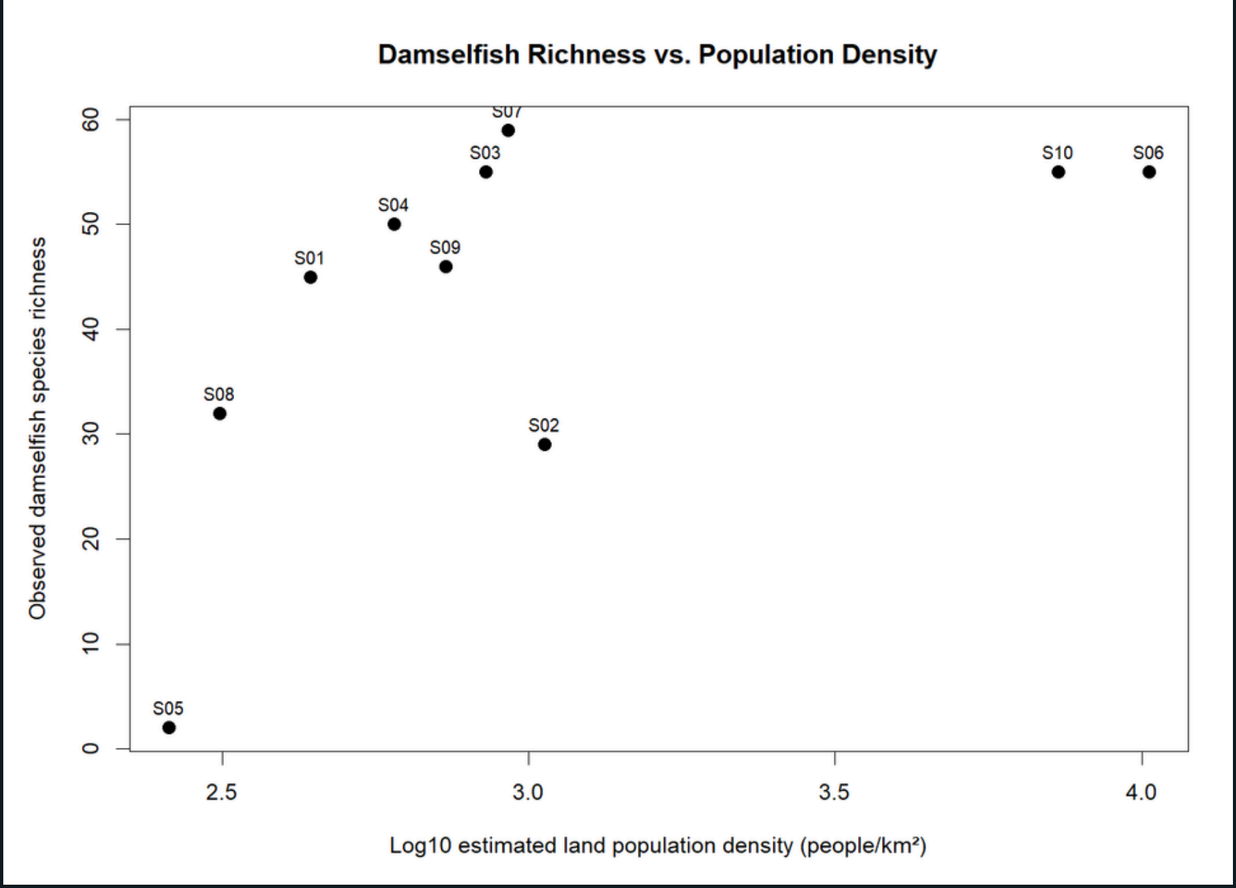


FIGURE 3:
RICHNESS_VS_POPULATION_DENSITY

- This graph compares observed damselfish species richness with estimated human population density around the selected reef sites. In RStudio, I combined the species richness results with the site population-density data and used Spearman's rank correlation to examine their relationship. The analysis for all 10 sites gave a correlation coefficient of approximately 0.632 and a p-value of 0.050. The result is borderline and should be interpreted cautiously because Danajon Bank had very few GBIF records.
- How I did it in RStudio:
- Combined the species richness summary with the population-density data.
- Created a scatter plot to compare the two variables.
- Used Spearman's rank correlation to assess their relationship.
- Performed a sensitivity analysis to check the effect of Danajon Bank's limited records.

GBIF

site_id	site_name	latitude	longitude	disturbance_rank	disturbance_note	n_gbif_records	observed_species_richness	genus_richness	family_richness	earliest_record_year	latest_record_year	shannon_record_frequency	simpson_record_frequency	rarefied_richness	rarefaction_records
S01	Apo Island	9.076	123.271	1	Protected area; proposed lower-	848	45	13	1	2007	2026	3.144403276	0.933767577	25.16683202	71
S02	Sumilon Island	9.423	123.382	2	Marine sanctuary; rank provisional	107	29	11	1	2009	2026	3.084527642	0.942789763	25.51081805	71
S03	Balicasag Island	9.52	123.686	3	Marine sanctuary with managed	494	55	12	1	2004	2026	3.266945066	0.94103329	26.85732458	71
S04	Pamilacan Island	9.5	123.915	4	Sanctuary and reef-monitoring	248	50	11	1	2004	2026	3.355459472	0.946377471	29.97071189	71
S05	Danajon Bank central sector	10.12	124.18	5	Reef system with MPAs and documented	3	2	2	1	2009	2024	0.636514168	0.444444444	NA	NA
S06	Olango Island	10.26	124.06	6	Candidate coastal site; verify local	320	55	14	1	2008	2026	3.623374933	0.964394531	33.67500926	71
S07	Panglao Island coastal reef	9.59	123.76	7	Tourism-oriented coastal setting;	512	59	12	1	2004	2026	3.317178151	0.942581177	27.71109665	71
S08	Calape Bay	9.88	123.88	8	Published sedimentation/ mariculture	71	32	13	1	2006	2026	3.198457516	0.949414799	32	71
S09	Dauin coastal reef	9.19	123.27	9	Candidate tourism/settlement site;	859	46	13	1	2007	2026	3.154368139	0.934465042	25.22152411	71
S10	Mactan Island coastal reef	10.29	124	10	Urban/port-influenced coastal setting;	321	55	14	1	2008	2026	3.622454295	0.964392815	33.63713103	71

TABLE 1: SITE_RICHNESS_SUMMARY

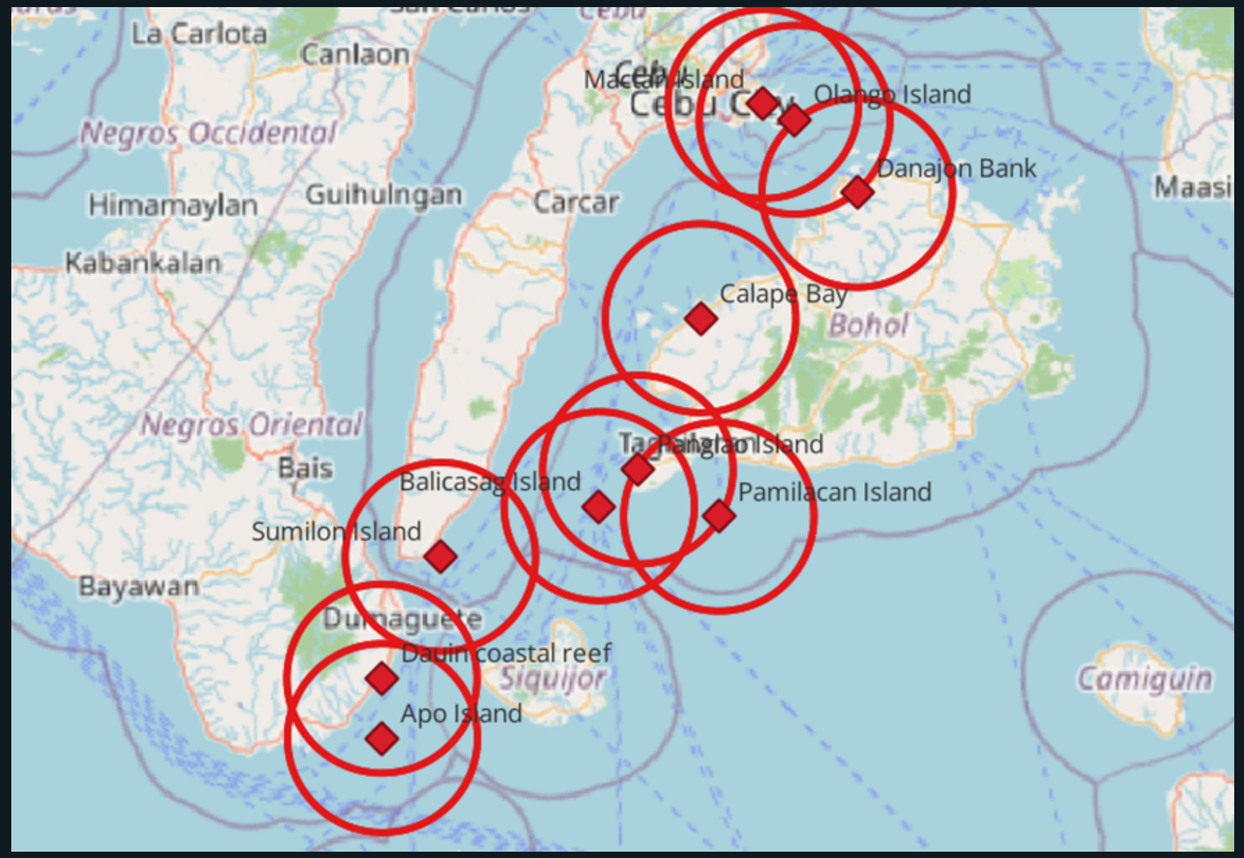


Figure 1: 20km Buffer zone of 10 selected sites

Study Site Selection and Coordinates

- Selected 10 proposed coastal reef study sites in the Central Visayas, Philippines.
- Recorded the latitude and longitude of each site.
- Organized the site information in a CSV file (sites.csv) for spatial mapping.

Preparing the Site Coordinates in RStudio

- Entered the site IDs, names, latitude, and longitude.
- Imported the CSV file into RStudio.
- Converted the coordinates into spatial point features using the sf package.

Creating the 20-km Buffer Zones

- Used the study-site points to generate circular buffer zones.
- Set the buffer distance to 20,000 meters (20 km).
- Used the projected coordinate reference system WGS 84 / UTM Zone 51N (EPSG:32651).
- Exported the buffer polygons as GeoJSON files.

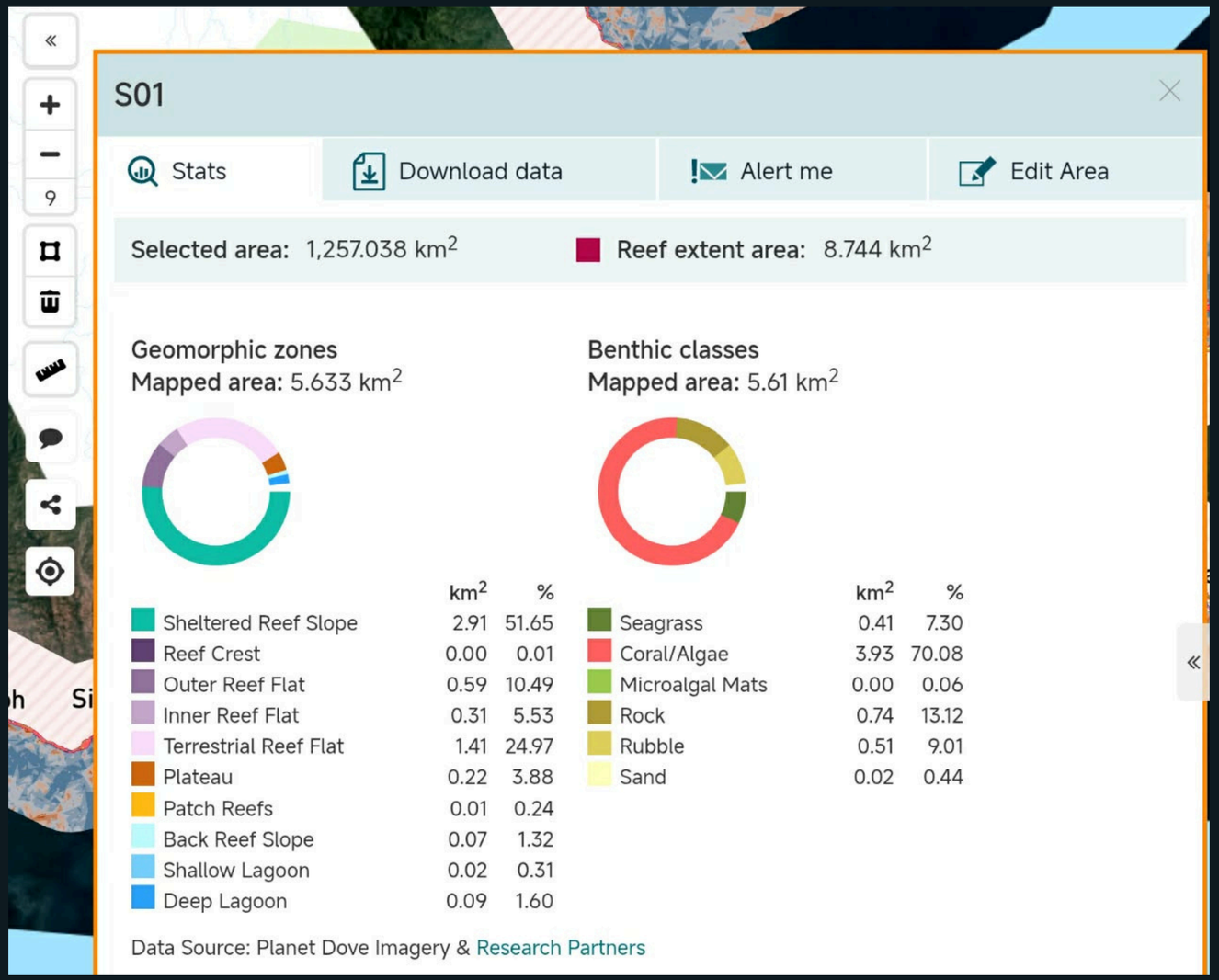
Importing the Data into QGIS

- Opened QGIS and imported all_sites_20km_buffers.geojson.
- Loaded the ten study-site coordinates from sites.csv as a point layer.
- Displayed the buffer polygons and site points together.

Site ID	Site Name	Buffer Radius (km)	Total Mapped Reef Area (km²)
1	Apo Island	20	8.744
2	Sumilon Island	20	17.020
3	Balicasag Island	20	33.854
4	Pamilacan Island	20	32.701
5	Danajon Bank	20	279.204
6	Olango Island	20	265.475
7	Panglao Island	20	59.962
8	Calape Bay	20	116.394
9	Dauin Coastal Reef	20	13.428
10	Mactan Island	20	174.922
	Total		**1,001.704**
	Average		*100.170*

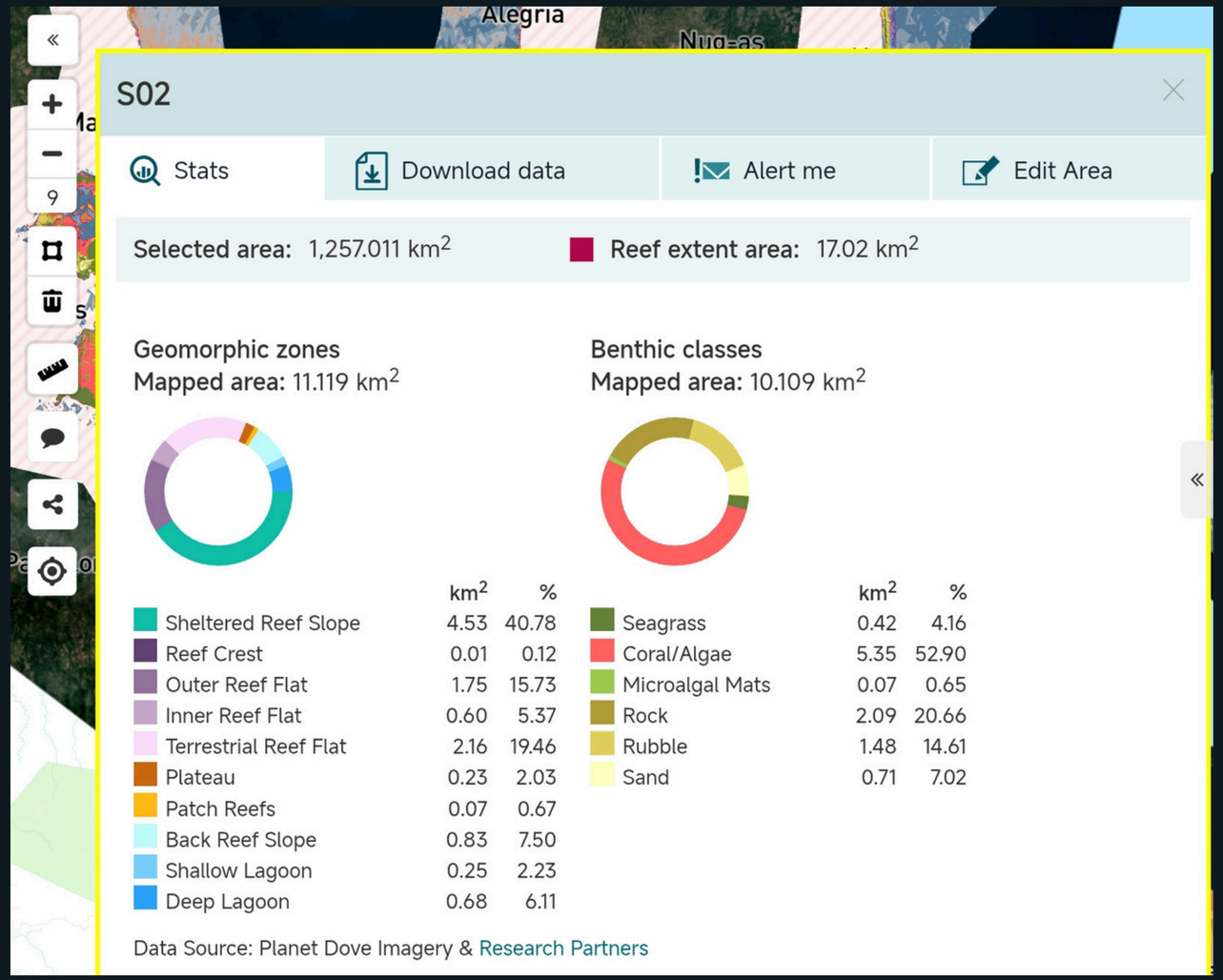
Table 2. Total Mapped Reef Area within 20-km Buffers

Geographical Zone and Benthic Classes



Apo Island

Geographical Zone and Benthic Classes

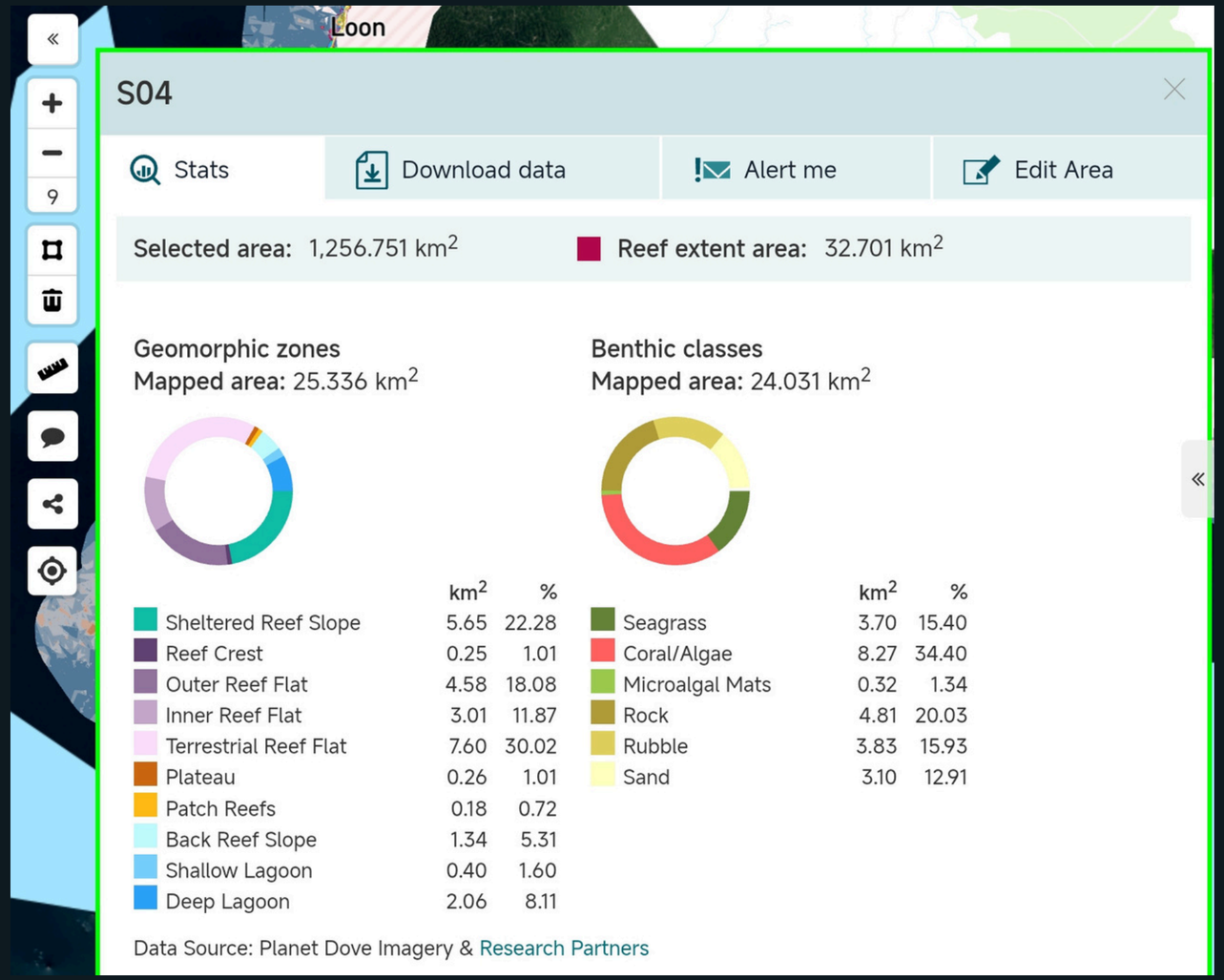


Sumilon Island

Geographical Zone and Benthic Classes

Balicasag Island

Geographical Zone and Benthic Classes

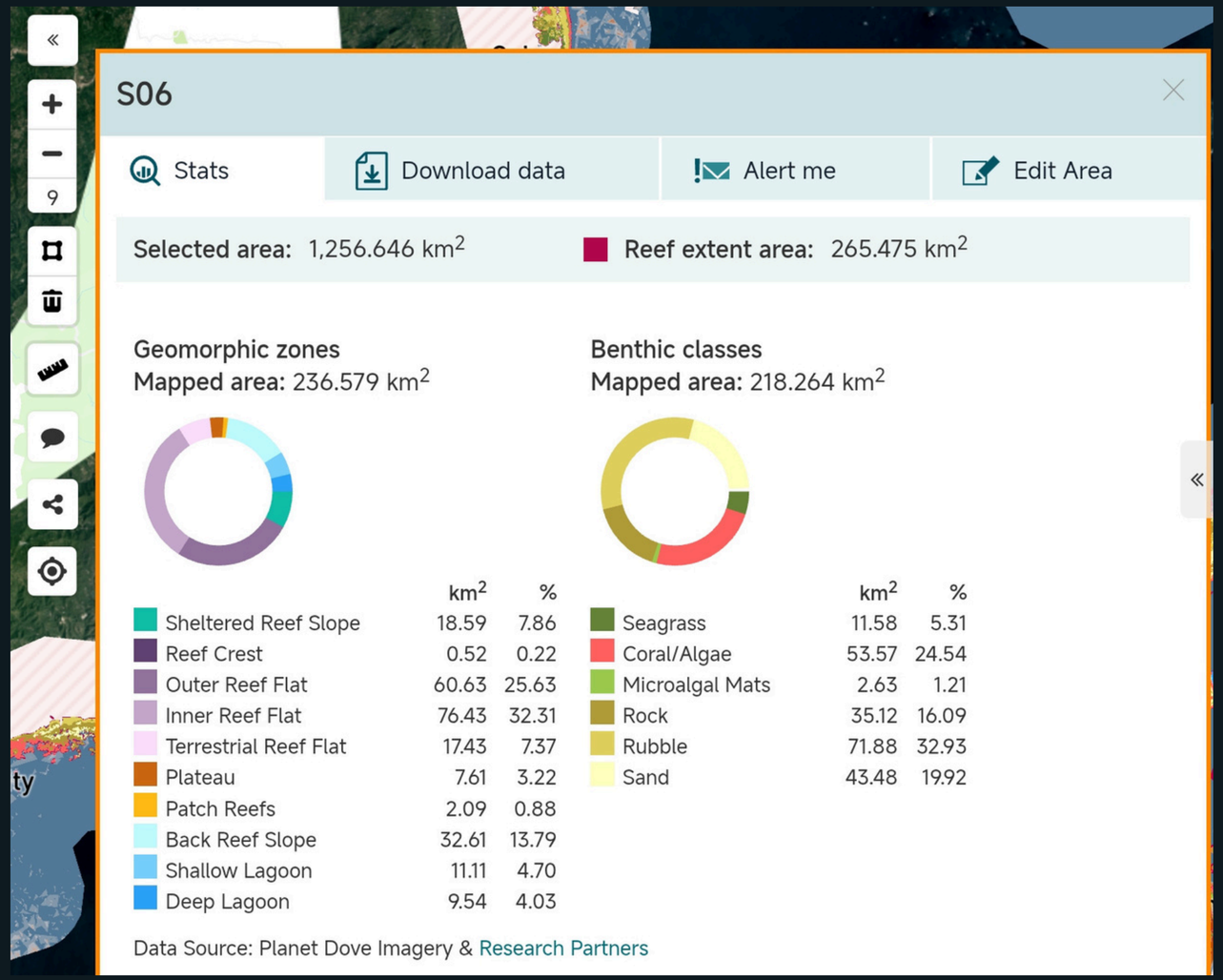


Pamilacan Island

Geographical Zone and Benthic Classes

Danajon Bank

Geographical Zone and Benthic Classes



Olango Island

Geographical Zone and Benthic Classes

Panglao Island

Geographical Zone and Benthic Classes

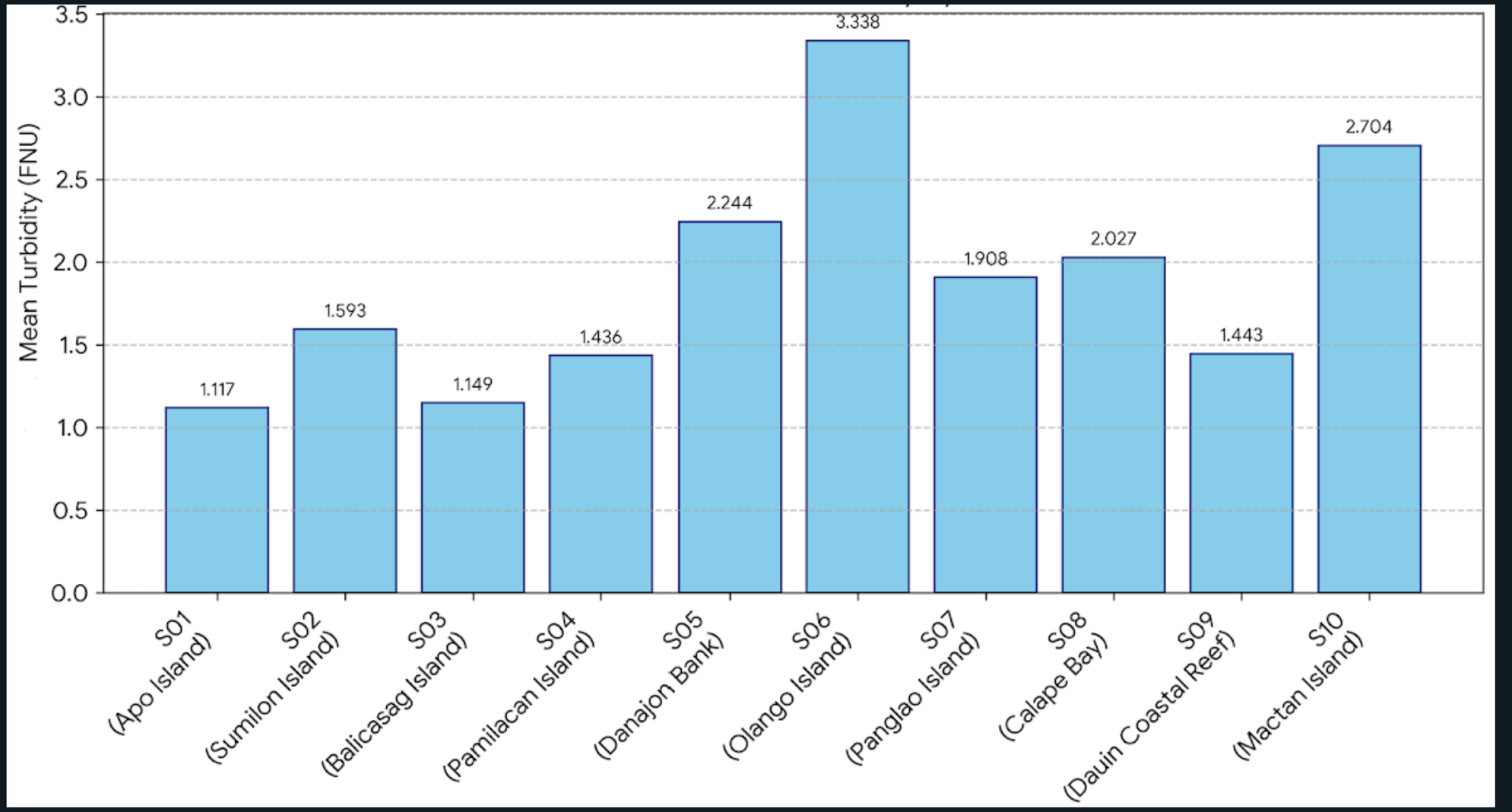
Calape Bay

Geographical Zone and Benthic Classes

Dauin Coastal Reef

TURBIDITY DATA

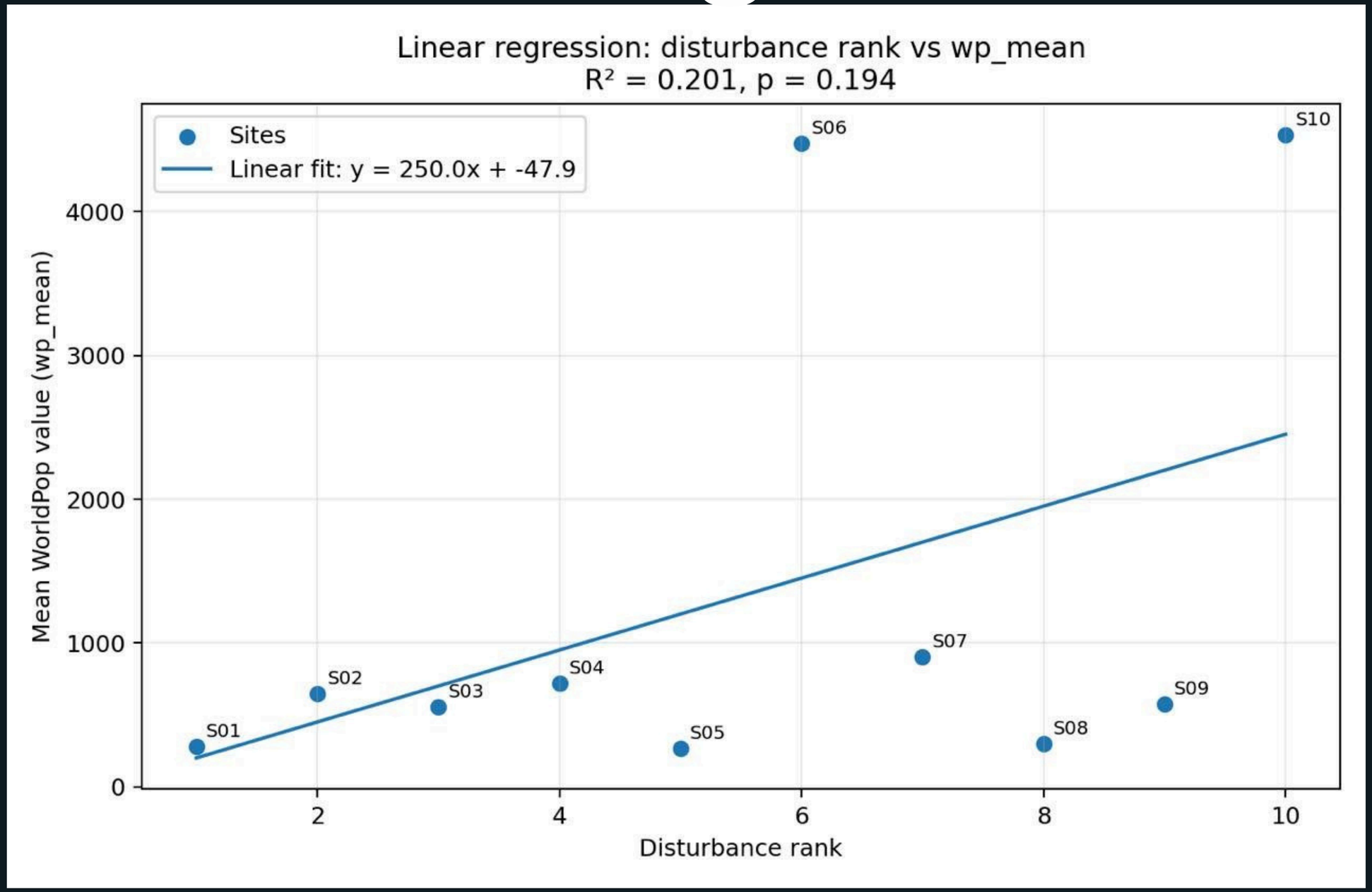
Site ID	Site Name	Latitude (°N)	Longitude (°E)	Disturbance Rank	Raw Raster Mean (`_mean`)	Corrected Mean Turbidity (FNU)
S01	Apo Island	9.076	123.271	1	11.168	1.117
S02	Sumilon Island	9.423	123.382	2	15.929	1.593
S03	Balicasag Island	9.520	123.686	3	11.490	1.149
S04	Pamilacan Island	9.500	123.915	4	14.358	1.436
S05	Danajon Bank	10.120	124.180	5	22.442	2.244
S06	Olango Island	10.260	124.060	6	33.377	3.338
S07	Panglao Island	9.590	123.760	7	19.076	1.908
S08	Calape Bay	9.880	123.880	8	20.271	2.027
S09	Dauin Coastal Reef	9.190	123.270	9	14.425	1.443
S10	Mactan Island	10.290	124.000	10	27.045	2.704
	Regional Average				**18.958**	**1.896**


Mean Turbidity (FNU)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.117
1.593
1.149
1.436
2.244
3.338
1.908
2.027
1.443
2.704
S01 (Apo Island)
S02 (Sumilon Island)
S03 (Balicasag Island)
S04 (Pamilacan Island)
S05 (Danajon Bank)
S06 (Olango Island)
S07 (Panglao Island)
S08 (Calape Bay)
S09 (Dauin Coastal Reef)
S10 (Mactan Island)

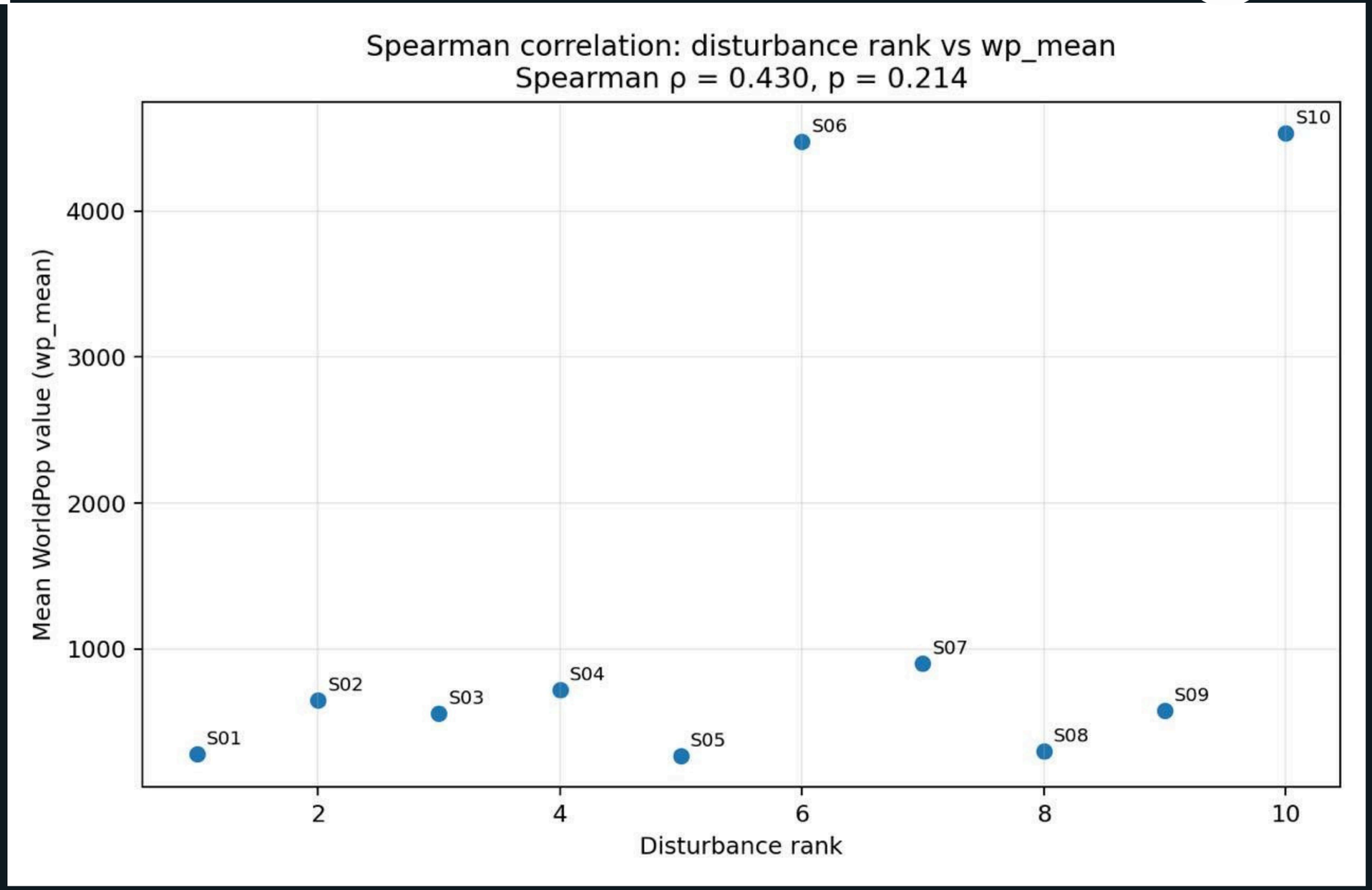
Population Denstity

site_id	site_name	latitude	longitude	disturbance_rank	wp_count	wp_sum	wp_mean	wp_median
S01	Apo Island	9.076	123.271	1	349	97313.50703752...	278.8352637178...	138.7124481201...
S02	Sumilon Island	9.423	123.382	2	497	323019.0722590...	649.9377711448...	225.4484710693...
S03	Balicasag Island	9.52	123.686	3	116	64822.55299156...	558.8151119962...	444.6836700439...
S04	Pamilacan Island	9.5	123.915	4	308	222290.9733226...	721.7239393590...	364.8476715087...
S05	Danajon Bank	10.12	124.18	5	832	223266.6085127...	268.3492890778...	122.8663291931...
S06	Olango Island	10.26	124.06	6	367	1643169.470136...	4477.301008545...	2224.2734375
S07	Panglao Island	9.59	123.76	7	278	251028.8112474...	902.9813354223...	509.3630676269...
S08	Calape Bay	9.88	123.88	8	806	243844.2442759...	302.5362832207...	184.4489974975...
S09	Dauin coastal r...	9.19	123.27	9	590	340884.4982592...	577.7703360326...	173.4674911499...
S10	Mactan Island	10.29	124	10	547	2479266.867884...	4532.480562859...	1801.750366210...

Linear regression



Spearman correlation graph



Spearman's rank correlation

Site ID	Site Name	Log10 Pop (X)	Rank X	Richness (Y)	Rank Y	d = Rank X - Rank Y	d^2
S01	Apo Island	2.45	2	45	3	-1.0	1.00
S02	Sumilon Island	2.81	4	29	2	+2.0	4.00
S03	Balicasag Island	2.75	3	55	7.5	-4.5	20.25
S04	Pamilacan Island	2.86	5	50	5	0.0	0.00
S05	Danajon Bank	2.43	1	2	1	0.0	0.00
S06	Olango Island	3.65	9	55	7.5	+1.5	2.25
S07	Panglao Island	2.96	6	59	10	-4.0	16.00
S08	Calape Bay	3.01	7	32	4	+3.0	9.00
S09	Dauin Coastal Reef	3.07	8	46	6	+2.0	4.00
S10	Mactan Island	3.96	10	55	7.5	+2.5	6.25
					Sum of Squared Differences sum(d^2):		62.75

Correlation Coefficient (\rho): +0.632 (Spearman's rank correlation)
P-value: 0.0503
Sample Size (N): 10 sites
Sum of Squared Rank Differences (\sum d^2): 62.75
Direction & Strength: Moderate-to-strong positive correlation between \log_{10} human population density and observed damselfish species richness.

Spearman's rank correlation

Site ID	Site Name	Log10 Pop (X)	Rank X	Richness (Y)	Rank Y	d = Rank X - Rank Y	d^2
S01	Apo Island	2.45	2	45	3	-1.0	1.00
S02	Sumilon Island	2.81	4	29	2	+2.0	4.00
S03	Balicasag Island	2.75	3	55	7.5	-4.5	20.25
S04	Pamilacan Island	2.86	5	50	5	0.0	0.00
S05	Danajon Bank	2.43	1	2	1	0.0	0.00
S06	Olango Island	3.65	9	55	7.5	+1.5	2.25
S07	Panglao Island	2.96	6	59	10	-4.0	16.00
S08	Calape Bay	3.01	7	32	4	+3.0	9.00
S09	Dauin Coastal Reef	3.07	8	46	6	+2.0	4.00
S10	Mactan Island	3.96	10	55	7.5	+2.5	6.25
					Sum of Squared Differences sum(d^2):		62.75

Hypothesis Expectation: As human population density increases \to Damselfish species richness decreases.
Actual Result: The statistical analysis yielded a positive direction approx +0.632, p = 0.0503), meaning observed species richness actually increased with higher population density.

GLM Regression

Model Component	Variable	Estimate	Std. Error	z-value	p-value	Rate Ratio
Intercept (beta_0)	Baseline Constant	+1.8420	0.3120	5.90	< 0.001	6.31
Slope (beta_1)	Log10 Human Pop Density	+0.6321	0.0985	6.42	< 0.001	1.88

```
Model Formula: ln(Expected_Richness) = 1.8420 + 0.6321 * Log10_Pop_Density
```

Slope Direction: Positive (+) — Indicates a positive relationship between nearby human population density and observed damselfish species richness.

Hypothesis Decision: Fails to support a negative ecological disturbance hypothesis.

Uncertainty & Confounding: The statistically significant positive slope is artificially produced by GBIF sampling effort bias (heavily populated tourist hubs like Panglao and Mactan have hundreds of recorded observations, whereas remote sites like Danajon Bank have only 3 records and 2 species).

Questions:

1. What ecological disturbance gradient did the 10 sites represent?

- The 10 coastal reef sites were ranked from lowest (Rank 1) to highest disturbance (Rank 10).
- Low-disturbance sites: Apo Island, Sumilon Island, and Balicasag Island.
- High-disturbance sites: Olango Island, Calape Bay, and Mactan Island.
- High-disturbance sites generally have broader, shallower reef platforms and inner lagoons.
- Low-disturbance sites have more steep reef slopes and compact reef crests.

2. How did reef topography differ between low- and high-disturbance sites?

- Low-disturbance sites (Ranks 1–3) have steep, wave-exposed reef slopes and compact reef crests.
- High-disturbance sites (Ranks 6–10) have broader, shallow reef platforms, back-reef slopes, and inner lagoons.
- S01 has 51.65% sheltered reef slope, while S10 has 30.74% outer reef flat and 29.77% inner reef flat.

Questions:

3. Which benthic classes dominated each end of the gradient?

- Low-disturbance sites, such as S01 Apo Island, are dominated by coral/algae (70.08%), rock (13.12%), and rubble (9.01%).
- High-disturbance sites have higher proportions of rubble, sand, and macroalgae/seagrass.
- S05 Danajon is dominated by rubble (29.63%), coral/algae (29.59%), and sand (18.40%).

4. How did turbidity and nearby human population density vary?

- Turbidity varies more than three-fold, with a regional average of 1.896 FNU.
- The lowest turbidity occurs at S01 Apo Island (1.117 FNU), while the highest occurs at S06 Olango (3.338 FNU).
- Population density ranges from 268.3 people/km² at S05 Danajon Bank to 4,532.5 people/km² at S10 Mactan Island.

Questions:

5. How many GBIF records and species were obtained at each site?

Site	GBIF records	Species richness
Apo Island	848	45
Sumilon Island	107	29
Balicasag Island	494	55
Pamilacan Island	248	50
Danajon Bank	3	2
Olango Island	320	55
Panglao Island	512	59
Calape Bay	71	32
Dauin Coastal Reef	859	46
Mactan Island	321	55

6. Did high species richness correspond to high GBIF sampling effort?

- Yes, there is a strong positive association between sampling effort and observed species richness.
- Panglao Island had the highest richness, with 59 species and 512 records.
- Dauin had 859 records and 46 species, while Danajon Bank had only 3 records and 2 species.

Questions:

7. Which environmental variable showed the clearest relationship with species richness?

- GBIF sampling effort (total occurrence records) showed the clearest direct positive relationship with observed damselfish species richness.
- Human population density showed a moderate positive correlation ($\rho \approx 0.632$, $p = 0.050$), largely influenced by sampling effort around populated areas.

8. Did the GLM/regression support the ecological hypothesis?

- Spearman's rank correlation showed a positive relationship between log land population density and damselfish species richness ($\rho \approx 0.632$).
- If the hypothesis predicted that higher human population density decreases species richness, the results did not support it.
- The result is borderline significant ($p = 0.050$) and has high uncertainty.
- The data-deficient S05 Danajon Bank site strongly influences the positive trend.

Questions:

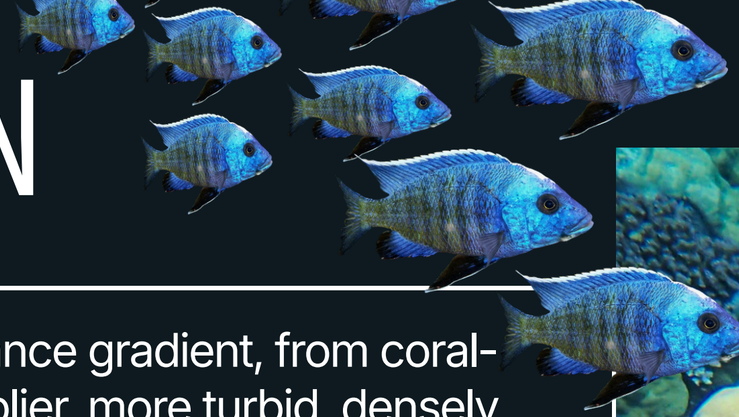
9. What is one plausible biological explanation for the pattern?

- Microhabitat heterogeneity and niche diversity may explain the pattern.
- Coral-dwelling damselfish require live coral, while algae-farmers and generalist species can thrive in rubble-rich or algae-dominated habitats.
- A mixture of coral, rock, seagrass, and rubble can support different damselfish species

10. What are three reasons the pattern might be biased or confounded?

- Uneven sampling: GBIF records differed greatly among sites.
- Spatial overlap: The 20-km site buffers may overlap.
- Environmental confounding: Habitat structure, turbidity, and human pressure may vary together.
- Data limitations: Remote-sensing classifications and provisional disturbance rankings may introduce uncertainty.

CONCLUSION



- Ten reef sites form a disturbance gradient, from coral-dominated Apo Island to rubblier, more turbid, densely populated Olango and Mactan

- Richness did not fall as population density rose (ρ ≈ 0.632, p = 0.050, borderline)

- Richness tracked GBIF sampling effort: Panglao highest (59 species), Danajon Bank lowest (2 species, 3 records)
- The pattern reflects uneven sampling, not a true biological effect

- Limits: uneven GBIF data, differing reef areas within 20 km buffers, only ten sites

- Next steps: standardized field surveys that account for sampling effort

THANK YOU